



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science Course

Code: DSA0404

Slot: A

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Submitted By

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Unit 3 Solved Problems

1. Mean, Median, Mode and Range

Data: 12, 15, 18, 15, 20, 22, 15, 25

Sorted: 12, 15, 15, 15, 18, 20, 22, 25

Mean = $142/8 = 17.75$

Median = $(15 + 18)/2 = 16.5$

Mode = 15

Range = $25 - 12 = 13$

2. Frequency Distribution

Data: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4

Value	Frequency
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2	3
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3	3
4	3
5	2
6	1
Total	12

3. Q1, Q2, Q3 and IQR

Data: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30

$$Q2 \text{ (Median)} = (15 + 18)/2 = 16.5$$

Lower half: 5, 8, 10, 12, 15 $\rightarrow Q1 = 10$

Upper half: 18, 20, 22, 25, 30 $\rightarrow Q3 = 22$

$$IQR = Q3 - Q1 = 22 - 10 = 12$$

Answer: $Q1 = 10$, $Q2 = 16.5$, $Q3 = 22$, $IQR = 12$

4. Outlier Detection Using IQR

Data: 10, 12, 13, 15, 16, 17, 18, 20, 45

$$Q1 = 13$$

$$Q3 = 20$$

$$IQR = 20 - 13 = 7$$

Lower limit:

$$13 - 1.5(7) = 2.5$$

Upper limit:

$$20 + 1.5(7) = 30.5$$

Since $45 > 30.5$, 45 is an outlier.

5. Skewness

Data: 5, 6, 7, 7, 8, 8, 9, 20

There is a large value of 20 on the right side.

Therefore, the distribution is:

Positively skewed (right skewed).

6. Uniform Distribution

Range = 10 to 50.

Formula:

$$P(a < X < b) = \frac{b-a}{50-10} \quad P(20 < X < 35) = \frac{35-20}{50-10} = \frac{15}{40} = 0.375$$

Answer = 0.375 or 37.5%

7. Z-Score

Given:

Mean = 100

SD = 15

X = 130

Formula:

$$Z = \frac{X - \mu}{\sigma} \quad Z = \frac{130 - 100}{15} = \frac{30}{15} = 2$$

Answer: Z = 2

8. Probability P (X < 130)

From Question 7:

$$Z = 2$$

From the standard normal table:

$$P(Z < 2) = 0.9772$$

$$\text{Answer} = 0.9772 \text{ or } 97.72\%$$

9. Effect of an Outlier

Example dataset:

10, 12, 14, 15, 16, 18, 100

Before treatment

$$\text{Mean} = 185/7 = 26.43$$

$$\text{Median} = 15$$

The outlier 100 greatly increases the mean.

After removing 100

Data:

10, 12, 14, 15, 16, 18

$$\text{Mean} = 85/6 = 14.17$$

$$\text{Median} = (14 + 15)/2 = 14.5$$

Conclusion

The mean changes significantly because it is sensitive to outliers, while the median changes much less.

10. Complete EDA – Student Marks

Example dataset:

45, 50, 55, 60, 62, 65, 68, 70, 72, 95

Central Tendency

$$\text{Mean} = 64.2$$

Median = 63.5

Mode = No mode

Dispersion

Minimum = 45

Maximum = 95

Range = $95 - 45 = 50$

Distribution

Most students scored between 45 and 72, while 95 is unusually high.

Outlier

Using the IQR method, 95 is not an outlier for this particular dataset.

Skewness

Since the higher value 95 pulls the mean upward:

Distribution is slightly positively skewed.

EDA Conclusion

The dataset has a mean of 64.2 and median of 63.5, indicating balanced central values. The range is 50, showing moderate variation, and the distribution is slightly right skewed.